



A Sustainable Tourism Web Platform for Pampanga.

Abstract

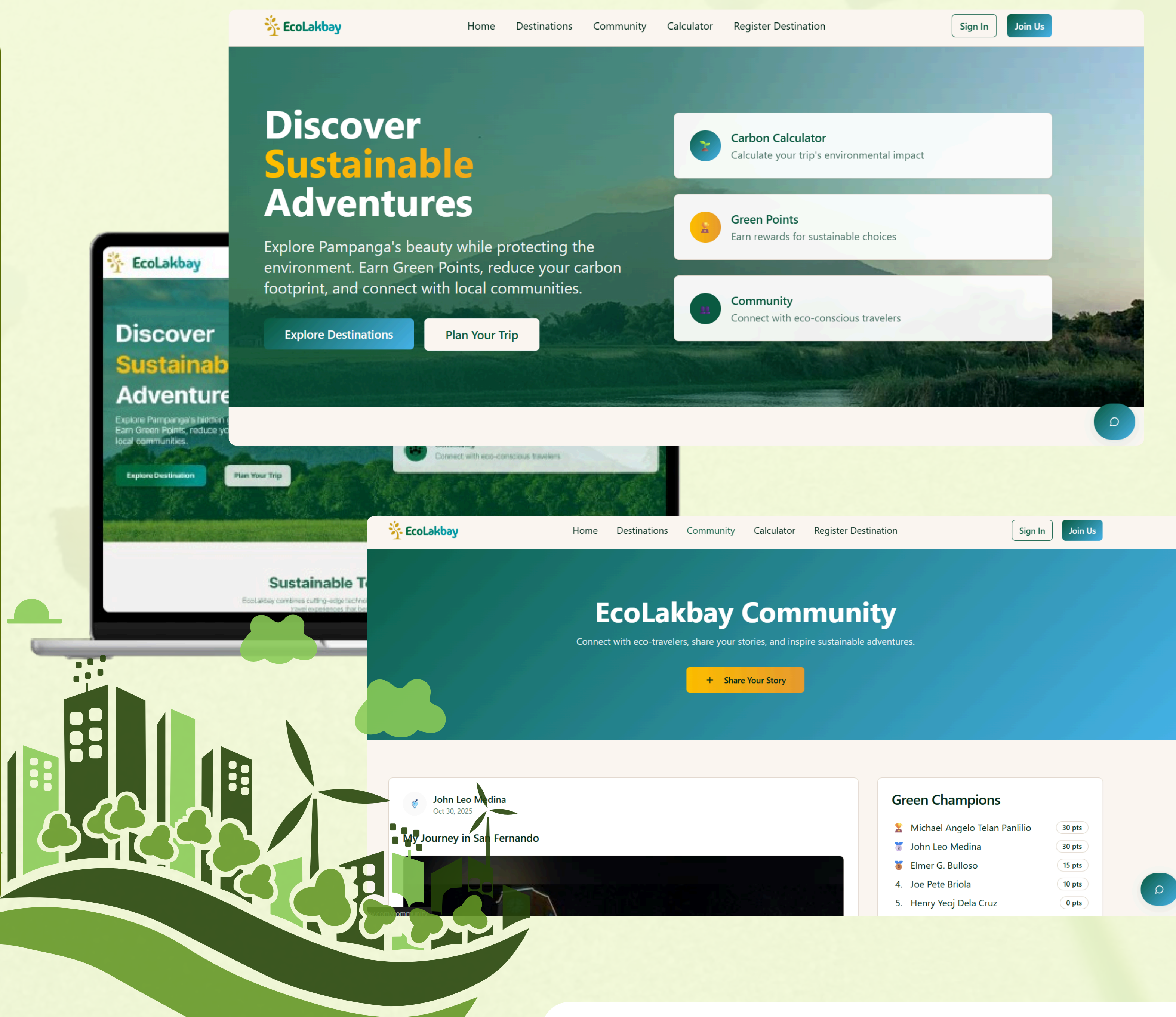
Tourism continues to accelerate economic growth in the Philippines, yet it also contributes to rising environmental issues such as waste generation, carbon emissions, and the degradation of natural and cultural sites, particularly in Pampanga. Addressing these challenges, this study developed EcoLakbay, a web-based sustainable tourism platform designed to guide travelers toward eco-conscious decisions through features such as a carbon footprint calculator, green points reward system, AI-assisted travel guidance, and community-driven destination reviews. Grounded in Sustainable Tourism Development principles, the Theory of Planned Behavior, and the Diffusion of Innovations, the study employed an exploratory sequential mixed-methods design, beginning with stakeholder interviews to identify sustainability gaps, which informed the system's design and iterative development using the Agile methodology. The platform was evaluated using the System Usability Scale (SUS) among tourists and stakeholders, and technically assessed through the ISO/IEC 25010 quality model by IT experts. Findings revealed key tourism challenges, including waste mismanagement, inconsistent traveler behavior, and limited coordinated sustainability practices, alongside a growing demand for environmentally responsible experiences. Usability scores indicated high acceptability, and expert evaluations affirmed the system's reliability and functional suitability. EcoLakbay demonstrates how digital tools can support responsible travel, strengthen stakeholder collaboration, and advance sustainable tourism in Pampanga.

Objectives

To discover how a digital travel platform can support the promotion and adoption of sustainable tourism practices in the Province of Pampanga by examining current environmental issues, traveler awareness and behavior, the role of stakeholders, and the potential impact of proposed features of EcoLakbay.

To design a user-friendly system that integrates key features such as a carbon footprint calculator, green points reward system, Destination review guides and Community Sharing wall to support eco-conscious travel decisions.

To evaluate the usability and effectiveness of the EcoLakbay platform through user testing among end users using the System Usability Scale (SUS) and expert validation using ISO/IEC 25010 software quality model.



Methods

Research Design:

Employs a mixed-methods exploratory sequential design, combining qualitative stakeholder interviews with quantitative system evaluations.

System Development:

Utilizes an Agile methodology (Scrum and Kanban) to iteratively build and refine the platform based on stakeholder user stories and ISO/IEC 25010 standards.

Participants & Setting:

Involves local tourism stakeholders, adult travelers (ages 18–60), and IT experts within the Province of Pampanga.

Research Instruments:

Data collection tools include semi-structured interview guides, the System Usability Scale (SUS) for user experience, and an ISO/IEC 25010 checklist for technical quality.

Data Analysis:

Uses thematic analysis for qualitative interview insights and descriptive statistics to measure the platform's usability, reliability, and security performance.

Results

- EcoLakbay achieved Very High Usability (SUS)
- High Software Quality (ISO/IEC 25010)
- Users showed strong willingness to use the system.
- Features effectively supported eco-friendly travel decisions
- Platform demonstrated readiness for real-world implementation

Discussion

EcoLakbay helps promote sustainable tourism by encouraging eco-friendly travel decisions and environmental awareness. The platform connects tourists, local communities, and tourism stakeholders, fostering shared responsibility for sustainability. Its strong usability and software quality results show that users can easily navigate and trust the system. The features guide travelers toward responsible choices while supporting local destinations. Overall, EcoLakbay serves as an effective digital tool for advancing sustainable and responsible tourism in Pampanga.